

Wikum Dinalankara, Ph.D.

Postdoctoral Associate, Weill Cornell Medicine — Department of Pathology and Laboratory Medicine
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Authorized to work in the United States — Permanent Resident.

PROFILE

Computational scientist focused on personalized prediction and outlier detection in high-dimensional biomedical data. PhD work introduced anti-profile / divergence methods that frame anomaly detection in omics as deviation from a learned baseline — a per-sample, n-of-1 framework directly applicable to personalized biomarker discovery. Subsequent postdoctoral work spans multi-omics oncology pipelines (genomics, transcriptomics), tumor heterogeneity quantification, and machine-learning pipelines for clinical flow cytometry. Strong publication record (h-index 15), open-source software development, and collaboration with clinicians and wet-lab biologists.

CORE COMPETENCIES

- **Personalized & outlier-detection methods** — rank-based divergence/anti-profile methods, integer-programming heterogeneity quantification, weakly-supervised classification.
- **Multi-omics integration** — paired DNA–RNA aberration analysis (PLOS Comp Bio); pan-cancer transcriptomic and methylation integration; cross-cohort feature harmonization.
- **Statistical & ML methods** — Bayesian / probabilistic modeling, regression and classification (LR, SVM, RF, XGBoost), CNNs for clinical flow cytometry, dimensionality reduction (UMAP, SOM), batch-effect correction, missing-value imputation.
- **Pathway & perturbation analysis** — differential-expression and signature-based pathway enrichment (KEGG, Reactome, MSigDB); immune-cell-state inference from bulk RNA-seq; perturbation profiling against baseline cohorts.
- **Programming & infrastructure** — Python (primary scientific stack: NumPy, pandas, scikit-learn, statsmodels, PyTorch); R/Bioconductor (package author); GCP / Vertex AI; SQL; Linux; Git; LaTeX.
- **Open-source software** — kTSP R/Bioconductor package (maintainer); divergence R package (lead author, PLOS ONE 2021); active GitHub at github.com/wikum.

EXPERIENCE

Postdoctoral Associate

September 2020 – Present

Weill Cornell Medicine — Department of Pathology and Laboratory Medicine, New York, NY

- Built ML pipelines on large clinical flow-cytometry datasets for B-cell-neoplasm classification (CNNs, random forests, UMAP/SOM-based methods); benchmarked accuracy and resource use across approaches (Cytometry Part B, 2024).
- Developed a weakly-supervised algorithm for chronic-lymphocytic-leukemia minimal-residual-disease detection that combines dimensionality reduction with random-forest classification on weakly-labeled flow data; preparing pilot deployment on Google Cloud Platform / Vertex AI.
- Multi-omic transcriptomic pipelines for collaborative oncology studies — Pan Prostate Cancer Group (DNA + RNA integration), Hodgkin lymphoma, bladder cancer — including Hodgkin/Reed–Sternberg cell transcriptomes (in review, Blood Cancer).
- Tumor immune-microenvironment characterization: comparative RNA-seq of tumor-infiltrating versus circulating lymphocytes (renal, bladder, prostate, glioblastoma) led to identification of a novel immunosuppressive cytokine, Meteorin-like (Immunity, 2024).
- Mesenchymal-state characterization in prostate cancer progression (Nature Communications, 2024) and prostate-cancer 3D-genome / multi-omic integration (ongoing).
- Translated complex analytical outputs into clinically-actionable summaries with pathologists, oncologists, and bioinformaticians; ensured all analyses complied with institutional data-governance and PHI policies.

Research Fellow

August 2015 – August 2020

The Johns Hopkins University — Department of Oncology, School of Medicine, Baltimore, MD

- Lead developer of the divergence / anti-profile framework: a rank-based, parameter-free outlier-detection method that converts gene-expression and methylation profiles into per-sample binary deviations from a learned normal-tissue baseline (PNAS 2018; PLOS ONE 2021). Directly applicable to personalized-biomarker discovery in heterogeneous patient populations.
- Co-led paired DNA–RNA aberration framework for multi-omic representation of tumor diversity across TCGA cancer types (PLOS Computational Biology 2021).

- Designed an integer-programming approach (Gurobi) to quantify per-cohort heterogeneity by selecting minimal feature sets that explain observed deviation patterns; used entropy-based metrics across cancer subtypes.
- Maintained and extended the ktSP R/Bioconductor package (rank-based, parameter-free classifier), adding feature-selection and applying it to clinical RNA-seq cohorts.
- Cross-domain regression modeling: epidemiological factors for influenza–COVID-19 cross-immunity; non-coding gene-expression analyses for the FANTOM consortium.

EDUCATION

Ph.D., Computer Science

August 2015

University of Maryland, College Park, MD — Dissertation: “Anti-profile methods for anomaly classification and regression”

B.Sc. (Hons.), Computer Science and Engineering

July 2008

University of Moratuwa, Sri Lanka — First-Class Honours, Dean’s List

SELECTED PUBLICATIONS

Full list: [Google Scholar \(T2E7-X4AAAAJ\)](#). * = co-first author.

- **Dinalankara W.**, Ng D.P., Marchionni L., Simonson P.D. “Comparison of three machine-learning algorithms for classification of B-cell neoplasms using clinical flow-cytometry data.” *Cytometry Part B: Clinical Cytometry*, 2024.
- Jackson C.M.*, Pant A.*, **Dinalankara W.***, Saleh L.*, et al. “The cytokine Meteorin-like inhibits anti-tumor CD8+ T-cell responses by disrupting mitochondrial function.” *Immunity*, 2024. (*co-first)
- Pakula H., Omar M., ... **Dinalankara W.**, ... Marchionni L., Loda M. “Distinct mesenchymal cell states mediate prostate-cancer progression.” *Nature Communications*, 2024.
- Ke Q.*, **Dinalankara W.***, Younes L., Marchionni L., Geman D. “Efficient representations of tumor diversity with paired DNA–RNA aberrations.” *PLOS Computational Biology*, 2021. (*co-first)
- **Dinalankara W.**, Ke Q., Geman D., Marchionni L. “An R package for divergence analysis of omics data.” *PLOS ONE*, 2021.
- **Dinalankara W.***, Ke Q.*, Xu Y., ... Younes L., Marchionni L., Geman D. “Digitizing omics profiles by divergence from a baseline.” *PNAS*, 2018. (*co-first)

SERVICE & RECOGNITION

- **Peer review** — Bioinformatics; PLOS Computational Biology; ISMB Proceedings; Journal of Data and Information Quality.
- **Teaching** — “Statistics and Data Analysis Using R” (Johns Hopkins, 2019); “Practical Genomics: Biology to Biostatistics” (Center for Computational Genomics, JHU, 2015–2017).
- **Memberships & awards** — American Association for Cancer Research (2016–present); Association for Pathology Informatics (2022–2025); Travel Award, API Summit 2023; Best Poster, JHU Symposium on Genomics & Bioinformatics.